

THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 7 (2020/2021)

1. Sketch the graph of each of the following:

- | | |
|----------------------|---|
| (a) i. $f(x) = 3^x$ | vi. $f(x) = \left(\frac{1}{4}\right)^x$ |
| ii. $f(x) = -3^x$ | vii. $f(x) = -\left(\frac{1}{4}\right)^x$ |
| iii. $f(x) = 4^{-x}$ | viii. $f(x) = 2^{x-1}$ |
| iv. $f(x) = 3^x + 2$ | ix. $f(x) = 2^{x+1}$ |
| v. $f(x) = 3^x - 2$ | |

(b) Solve each of the following equations:

- | | |
|--|--|
| i. $3^x = 81$ | iv. $2^x = 0.125$ |
| ii. $x^3 = 125$ | v. $27^{4x} = 9^{x+1}$ |
| iii. $\left(\frac{1}{2}\right)^x = \frac{1}{16}$ | vi. $\left(\frac{1}{8}\right)^{-2t} = 2^{t+3}$ |

(c) Solve each of the following equations:

- | | |
|------------------------------|------------------------------------|
| i. $2^{2x} + 3(2^x) - 4 = 0$ | iii. $4^x - 6(2^x) - 16 = 0$ |
| ii. $4^x - 2^{x+1} = 48$ | iv. $2^{2x+1} - 2^{x+1} + 1 = 2^x$ |

2. (a) Write each of the following in logarithm form:

- | | |
|--|----------------------|
| i. $y = 5^{x+2}$ | iii. $y = e^x$ |
| ii. $\left(\frac{3}{4}\right)^{-2} = \frac{16}{9}$ | iv. $8^{-1} = 0.125$ |

(b) Write each of the following in exponential form:

- | | |
|---|----------------------------------|
| i. $\log_2 32 = 5$ | iii. $\log_{27} 3 = \frac{1}{3}$ |
| ii. $\log_2 \left(\frac{1}{16}\right) = -4$ | |

(c) Evaluate each of the following:

i.

$$\log_3 9$$

ii.

$$\log_5 125$$

iii.

$$\log_6 \left(\frac{1}{36} \right)$$

3. (a) Solve each of the following equations:

i.

$$\log_3 x = 4$$

iii.

$$\log_a 2 = \frac{1}{2}$$

v.

$$\ln x = -2$$

ii.

$$\log_4 x = \frac{3}{2}$$

iv.

$$\log_2(3x + 4) = 4$$

(b) Solve each of the following equations:

i.

$$\log_3 y = 2 \log_3 7$$

iv.

$$\log_3(x^2 + 2) = 1 + \log_3(x + 2)$$

ii.

$$\log_5 x - \log_5 4 = 2$$

v.

iii.

$$\log_x 8 + \log_x 4 = 5$$

$$\log_4(x^2 + 8x - 1) = 2 + \log_4(x - 1)$$

(c) Sketch each of the following in the indicated interval:

i.

$$f(x) = \log_7 x$$

iii.

$$f(x) = -\log_3 x$$

ii.

$$f(x) = \log_{\frac{1}{7}} x$$

iv.

$$f(x) = \log_{\frac{1}{3}} x$$

(d) Sketch the graph of f and g on the same coordinate plane.

i.

$$f(x) = 3^x, \quad g(x) = \log_3 x$$

ii.

$$f(x) = \left(\frac{1}{2} \right)^x, \quad g(x) = \log_{\frac{1}{2}} x$$

iii.

$$f(x) = e^x, \quad g(x) = \ln x$$